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Question 4.17

"Which Theorems were used in finding the above equivalent ways to negate the proposition (4.1)?"

Theorem 4.15

Exercise 4.18

"Negate each of the following. Disregard the truth value."

(a) $(\forall n \in \mathbb{N})(\exists m \in \mathbb{N})(m < n)$.

$$\neg[(\forall n \in \mathbb{N})[(\exists m \in \mathbb{N})(m < n)]](\exists n \in \mathbb{N})\neg[(\exists m \in \mathbb{N})(m < n)](\exists n \in \mathbb{N})(\forall m \in \mathbb{N})\neg(m \geq n)$$

(b) $(\forall x, y, z \in \mathbb{Z})[(xy\text{is even and yz is even}) \implies xz \text{ is even}]$.

$$(\exists x, y, z \in \mathbb{Z})(\text{xy is even and yz is even} \wedge \text{xz is odd}).$$

(c) For all positive real numbers x, there exists a real number $y : y^2 = x$.

$$(\forall x \in \mathbb{R}^+)(\exists y \in \mathbb{R}) : (y^2 = x)(\exists x \in \mathbb{R}^+)(\forall y \in \mathbb{R}) : (y^2 \neq x)$$

(d) There exists a married person x such that for all married people y, x is married to y.

Let M be the set of married people. Let $x \heartsuit y$ denote that x is married to y.

$$(\exists x \in M) : (\forall y \in M)(x \heartsuit y).(\forall x \in M) : (\exists y \in M)\neg(x \heartsuit y).$$

Problem 4.21

(a) True. Consider $n = 1$.

(b) False. Consider the case where $y = x + 1$.

(c) False. Consider $x = -1$.

(d) True. Consider $m = -n$.

Proof.

We use proof by contradiction. Thus, assume that $\exists n \in \mathbb{Z} : \forall m \in \mathbb{Z}$, then $n + m \neq 0$. Consider the case where $m = -n$. Now $n - n = 0$, which contradicts our previous assumption. Thus, our previous assumption leads to a contradiction. Therefore $\forall n \in \mathbb{Z}$ there exists $m \in \mathbb{Z} : n + m = 0$.

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(e) True. Consider $a = -3, b = 1$.

Proof.

We use proof by contradiction. Thus, assume that $(\forall a, b)(2a + 7b \neq 1)$. Consider the case where $a = -3, b = 1$. Now $2a + 7b = 1$, which contradicts our previous assumption. Thus, our previous assumption leads to a contradiction. Therefore $(\exists a, b)(2a + 7b = 1)$.

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(f) True. We know $2m$ and $4n$ are always even, so their sum is even.

(g) True by Theorem.

Theorem 4.22

Proof.

We use proof by contradiction. Thus, assume that there exists an integer z such that for all natural numbers, n, then $4z \neq 1 + (-1)^n(2n - 1)$. Consider the case where $n = 2z$. Now $(-1)^{2z} = 1$, because $2z$ is even. Then $4z = 1 + 2n - 1 = 2n = 4z$ $, which is a contradiction. Thus assuming there exists an integer z such that for all natural numbers, n, then $4z \neq 1 + (-1)^n(2n - 1)$, leads to a contradiction. Therefore, for all integers z, then there exists a natural number, n, such that $4z = 1 + (-1)^n(2n - 1)$.

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Theorem 4.23

Proof.

We use proof by contradiction. Thus, assume that $\exists n : 3n^2 + n + 14$ is odd. There are two cases, where n is even or not.

If n is even, then, $n = 2k$ for some integer k. Thus, $3n^2 + n + 14$

$$= 3(2k)^2 + 2k + 14 = 12k^2 + 2k + 14 = 2(6k^2 + k + 7).$$

Now, we know the term $2(6k^2 + k + 7)$ is even, which contradictions our original assumption.

If n is odd, then $n = 2k + 1$ for some integer k. Thus, $3n^2 + n + 14$

$$= 3(2k + 1)^2 + 2k + 15 = 12k^2 + 14k + 18 = 2(6k^2 + 7k + 9).$$

Now, we know the term $2(6k^2 + 7k + 9)$ is even, which contradicts our original assumption.

So, assuming that $\exists n : 3n^2 + n + 14$ is odd always leads to a contradiction. Therefore, $\forall n : 3n^2 + n + 14$ is even.

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